

Photosynthesis and Chlorophyll Fluorescence: Principles, Regulation, and PAM Fluorometry

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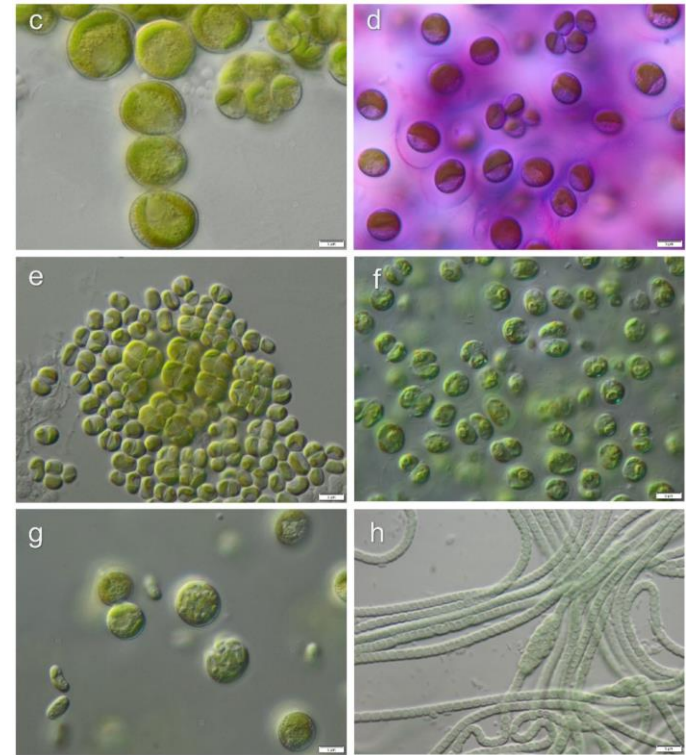
ljami@imedea.uib-csic.es & ludovic.jami@gmail.com

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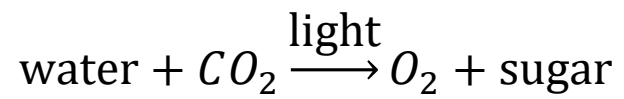
Higher plants

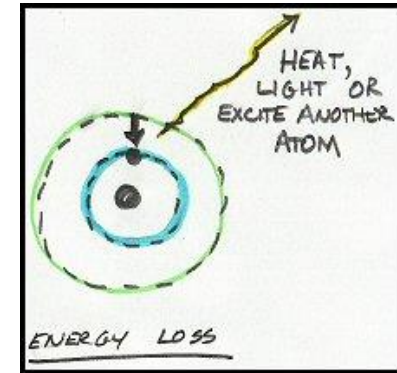
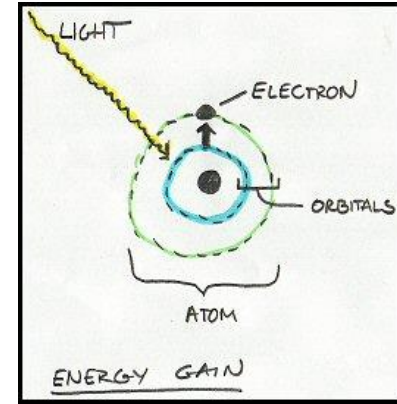
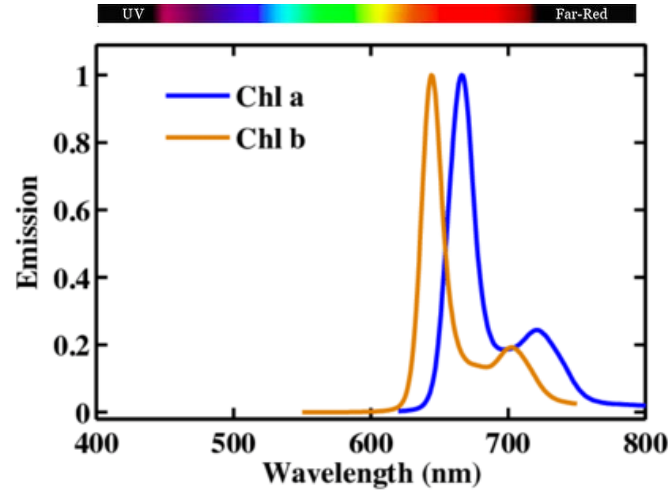
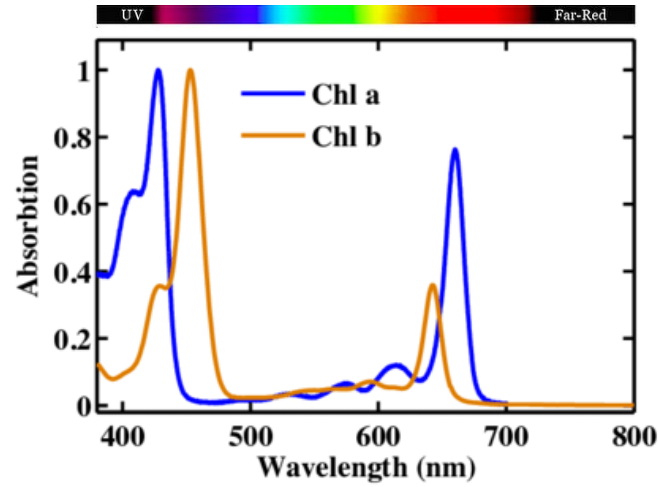


Algae and cyanobacteria



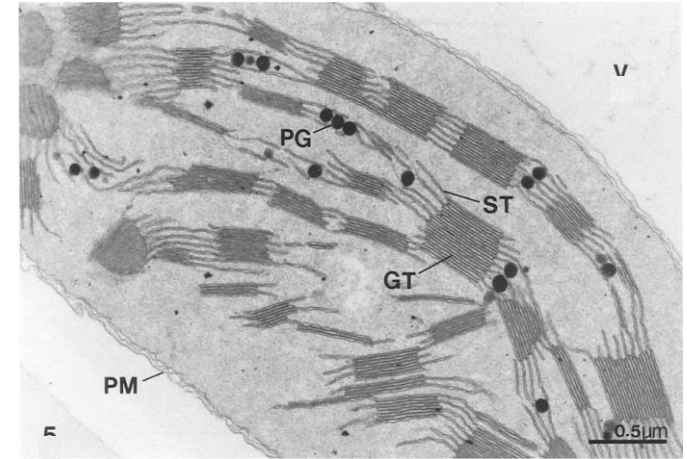
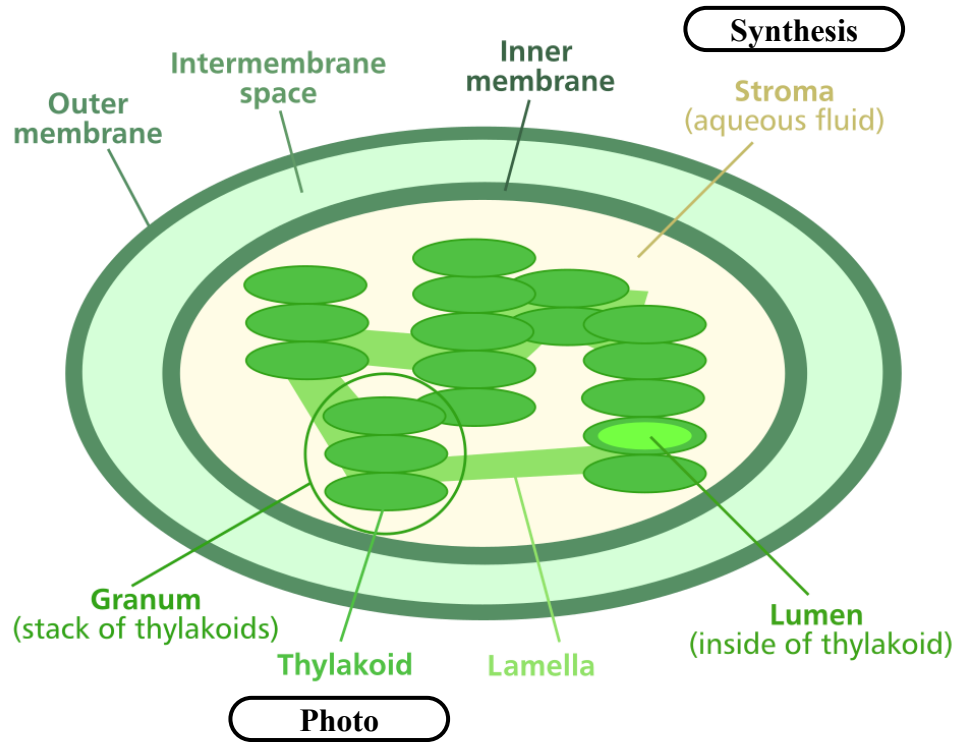
Samolov et al., Microorganisms, 2020



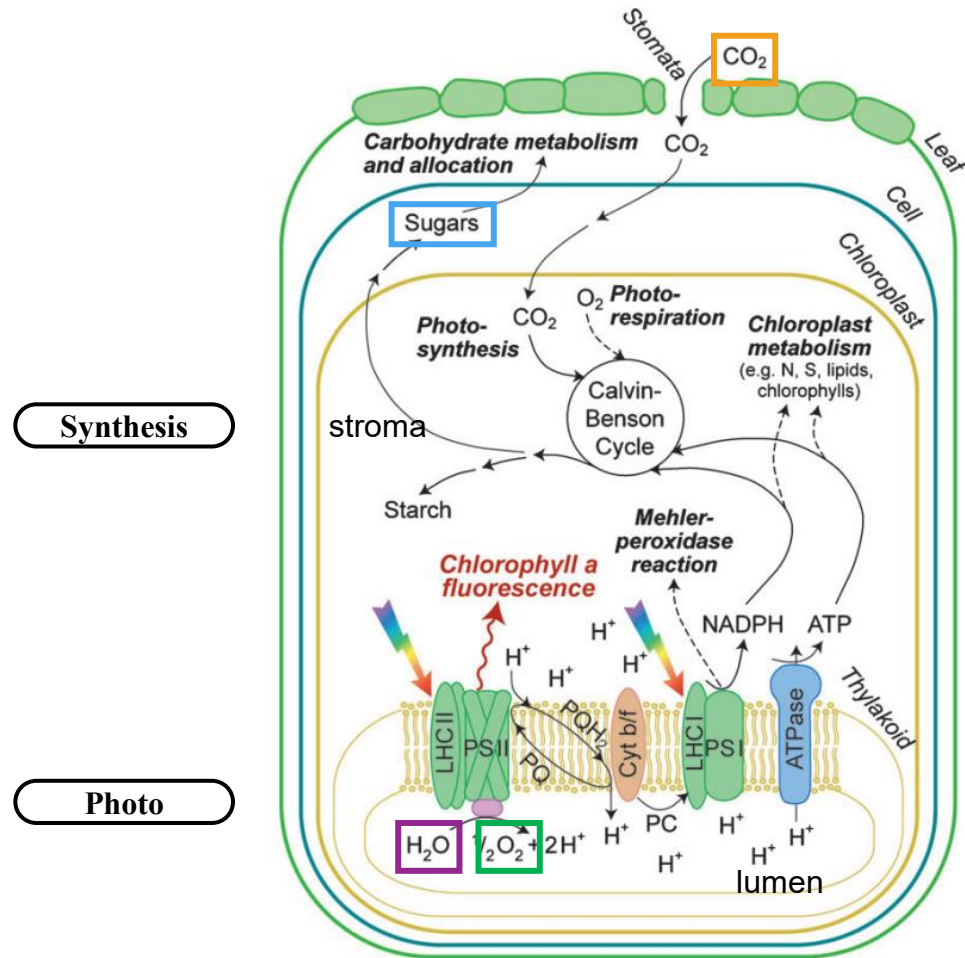


PhD Thesis Mammadov Gulmammad, 2015.

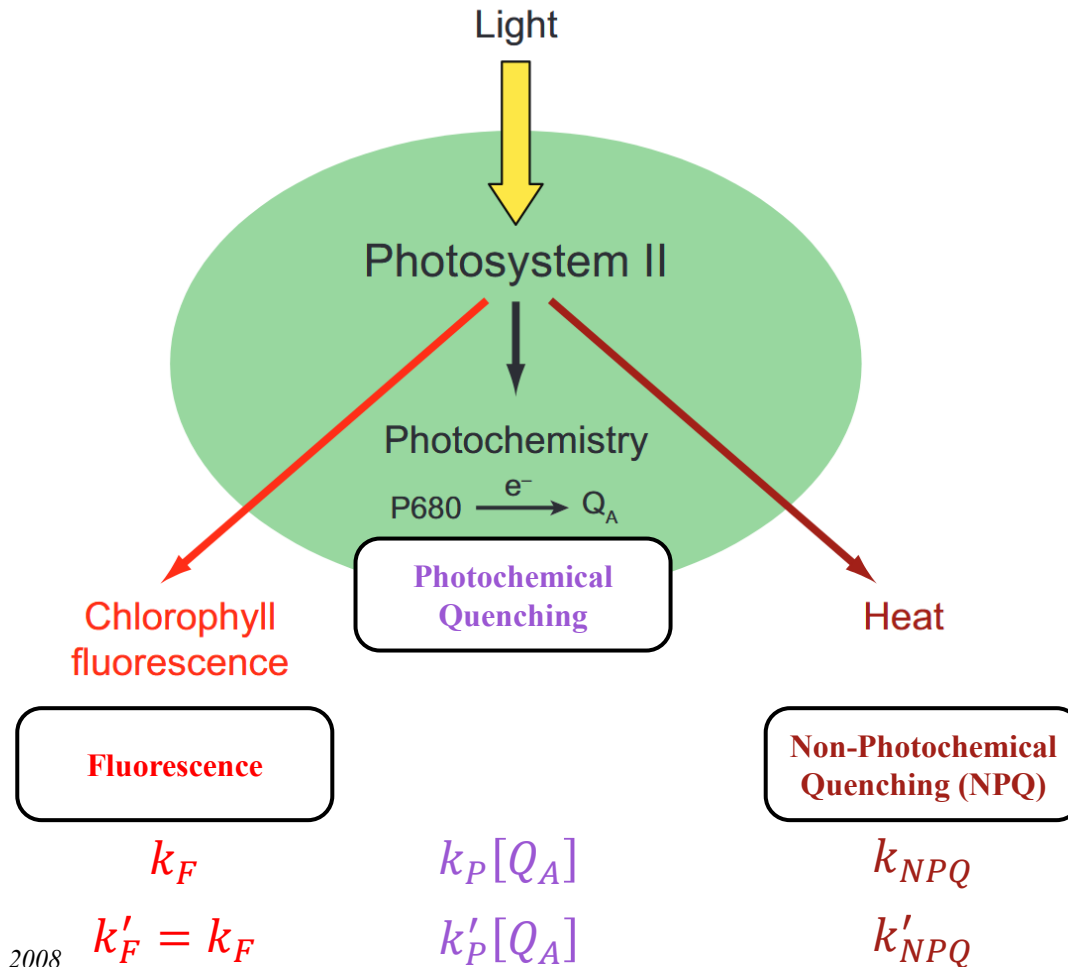
Energy gain and loss. Tom Michaels

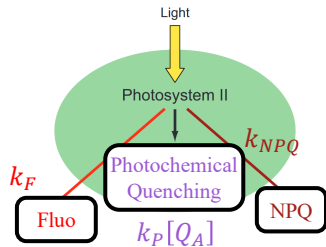


GT: grana thylakoids
 ST: stromal thylakoids
 PM: plasma Membrane
 PG: plastoglobuli



$[Q_A]$ fraction of oxidised quinone A





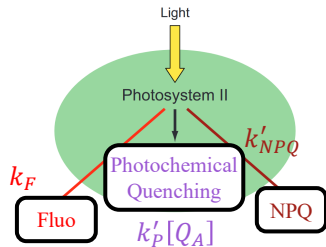
Dark-adapted

Fluorescence signal proportional to $\Phi_F = \frac{k_F}{k_F + k_P[Q_A] + k_{NPQ}}$

- $F_0 \propto \frac{k_F}{k_F + k_{NPQ} + k_P}$: fluorescence of dark adapted cells ($[Q_A] = 1$).
- $F_M \propto \frac{k_F}{k_F + k_{NPQ}}$: *maximum* fluorescence of dark adapted cells ($[Q_A] = 0$).
- $F_v = F_M - F_0$: variable fluorescence.

Maximum quantum yield of photosynthesis

$$\frac{F_v}{F_M} = \Phi_{PSII}^{\max} = \frac{k_P}{k_F + k_P + k_{NPQ}}$$



Light-adapted

- $F' \propto \frac{k_F}{k_F + k'_{NPQ} + k'_P[Q_A]}$: operating fluorescence in light conditions;
- $F'_0 \propto \frac{k_F}{k_F + k'_{NPQ} + k'_P}$: minimum fluorescence in light conditions;
- $F'_M \propto \frac{k_F}{k_F + k'_{NPQ}}$: maximum fluorescence in light conditions;

$F'_v = F'_M - F'_0$: variable fluorescence in light conditions

$F'_q = F'_M - F'$: quenched fluorescence in light conditions

$$\frac{F'_v}{F'_M} = \Phi_{PSII}^{\max}(I) = \frac{k'_P}{k_F + k'_P + k'_{NPQ}}$$

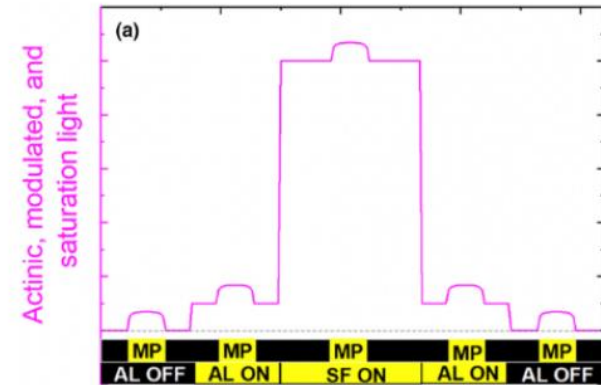
$$\frac{F'_q}{F'_M} = \Phi_{PSII}(I) = \frac{k'_P[Q_A]}{k_F + k'_P[Q_A] + k'_{NPQ}}$$

Absolute fluorescent signal is not required, relative changes are enough.

ML (or MP): Measuring Light (or Pulse), very weak, modulated light used to measure fluorescence without significantly driving photosynthesis

AL: Actinic Light, continuous illumination driving photosynthesis and physiological adaptation.

SP (or SF): Saturating Pulse (or Flash), brief and extremely intense flash that temporarily closes all PSII reaction centres



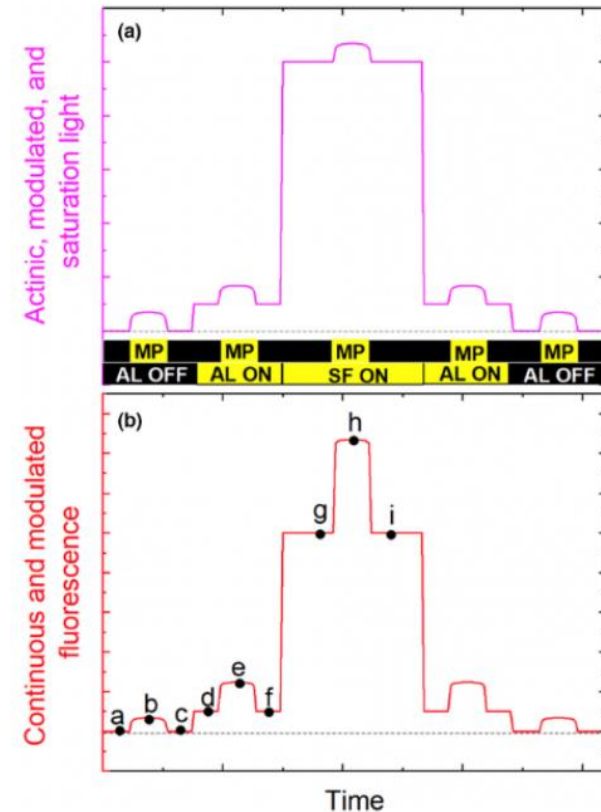
Demo on: Microscopy PAM

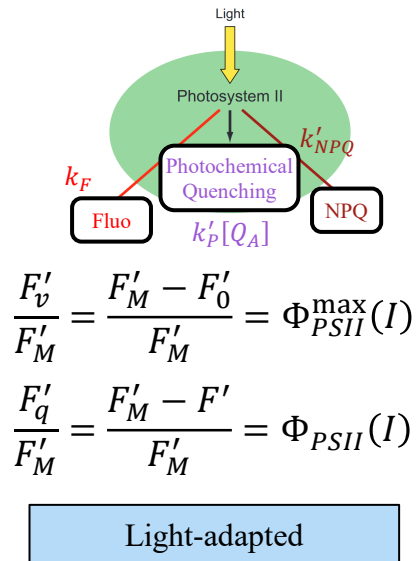
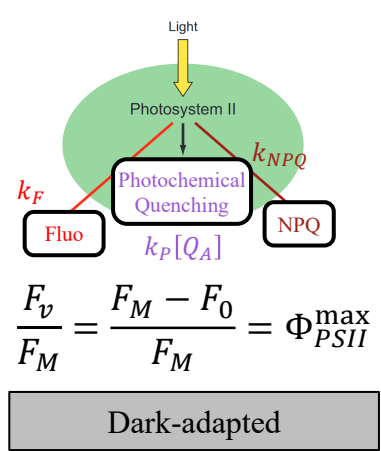
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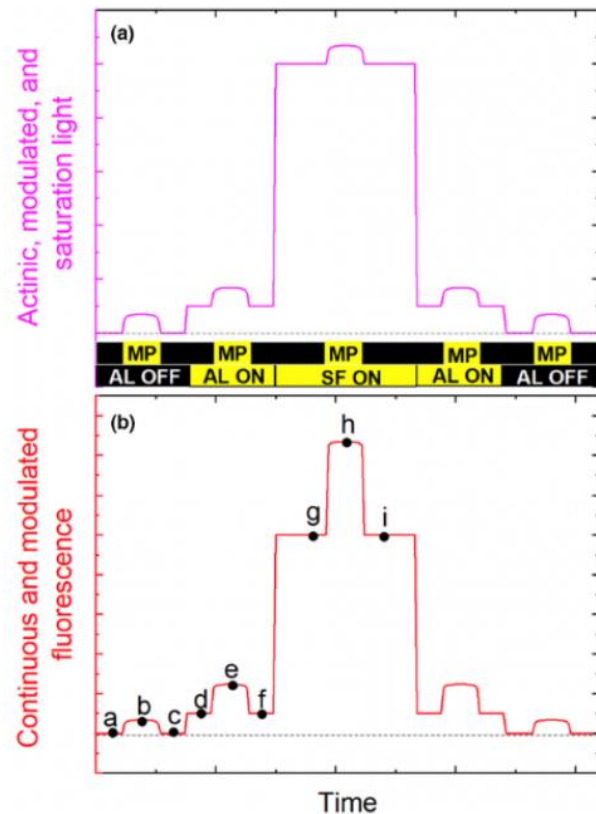


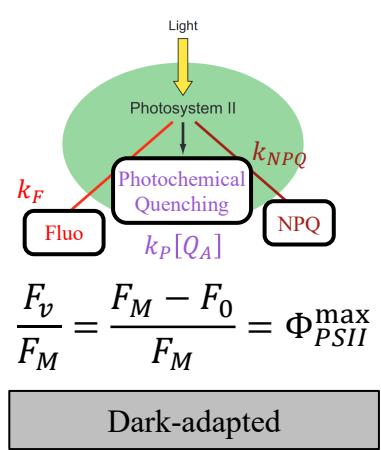


$$F_0 =$$

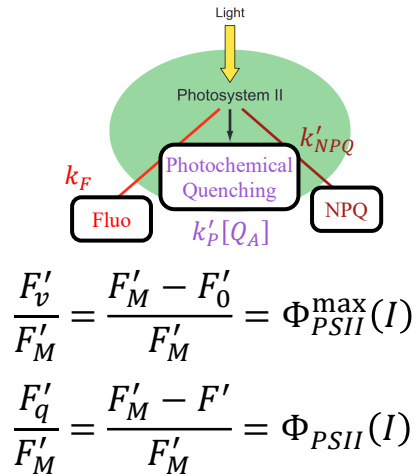
$$F'_0 =$$

$$F'_M =$$





$$\frac{F_v}{F_M} = \frac{F_M - F_0}{F_M} = \Phi_{PSII}^{\max}$$



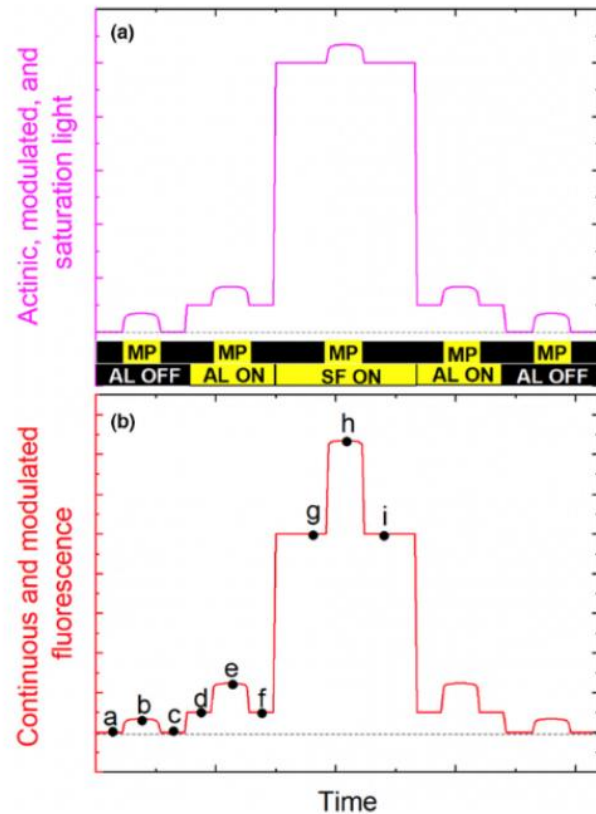
$$\frac{F'_v}{F'_M} = \frac{F'_M - F'_0}{F'_M} = \Phi_{PSII}^{\max}(I)$$

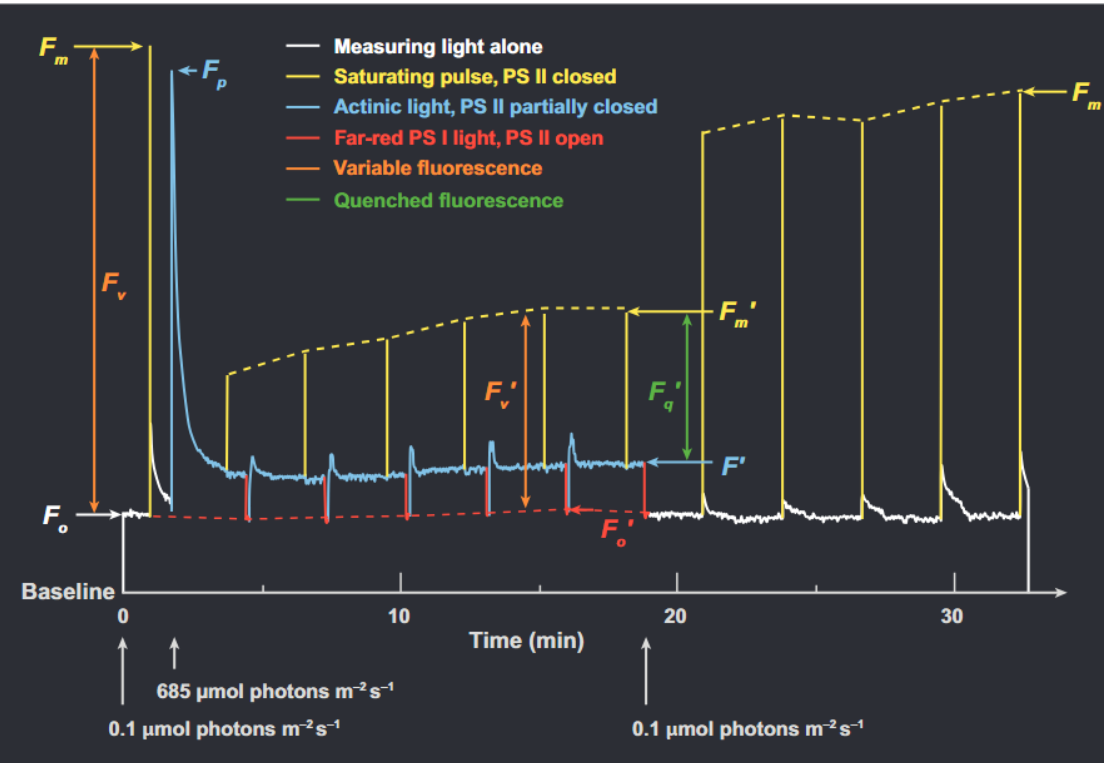
$$\frac{F'_q}{F'_M} = \frac{F'_M - F'}{F'_M} = \Phi_{PSII}(I)$$

$$F_0 = b - \frac{a + c}{2}$$

$$F'_0 = e - \frac{d + f}{2}$$

$$F'_M = h - \frac{g + i}{2}$$





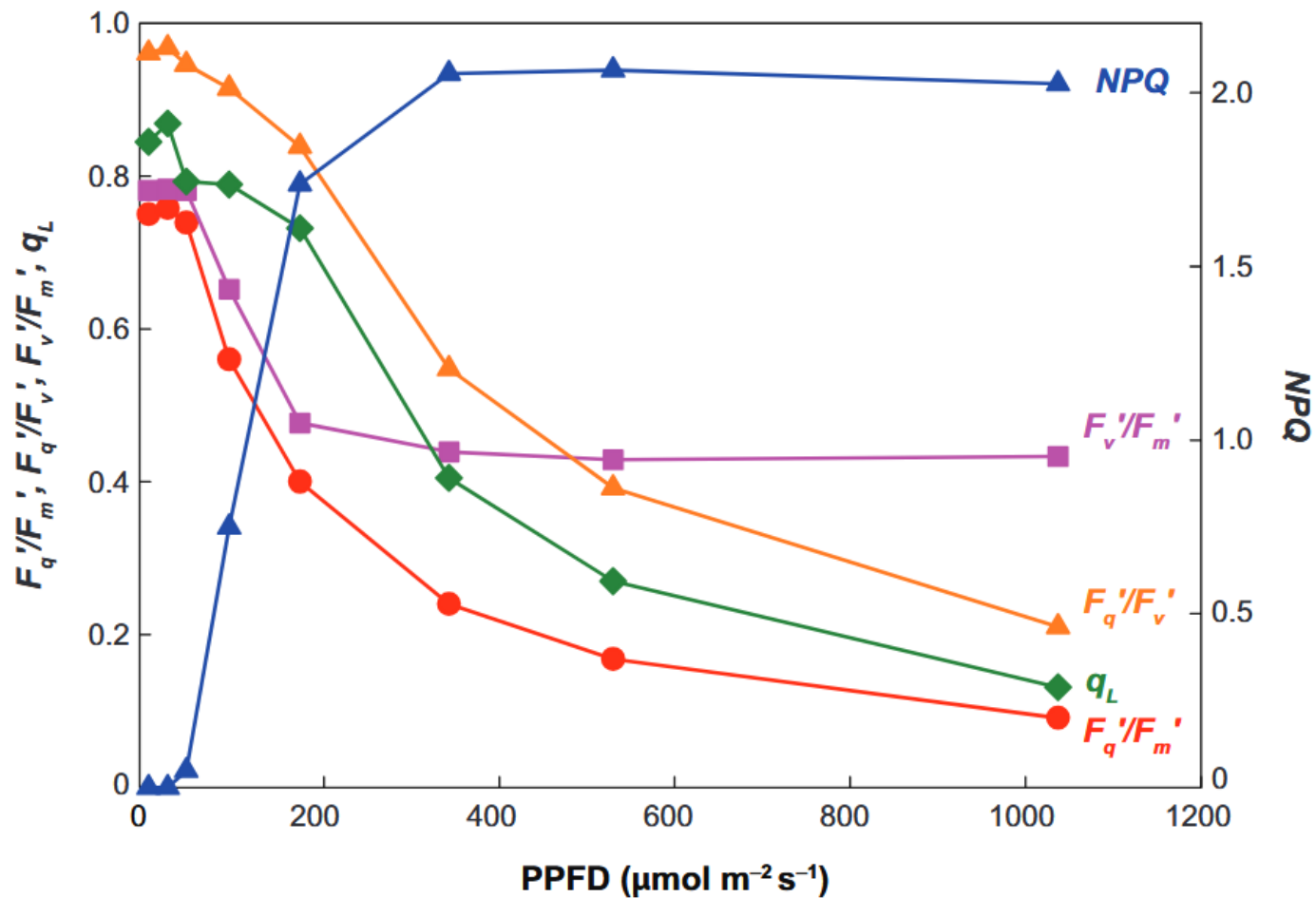
$$\frac{F_v}{F_M} = \frac{F_M - F_0}{F_M} = \Phi_{PSII}^{\max}$$

$$\frac{F'_v}{F'_M} = \frac{F'_M - F'_0}{F'_M} = \Phi_{PSII}^{\max}(I)$$

$$\frac{F'_q}{F'_M} = \frac{F'_M - F'}{F'_M} = \Phi_{PSII}(I)$$

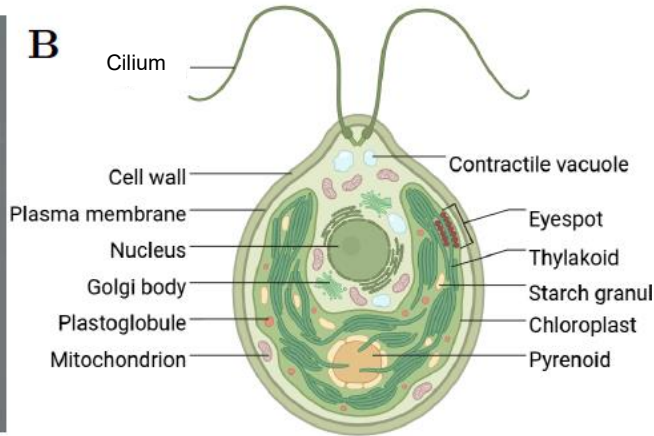
$$[Q_A] \text{ can be estimated as } [Q_A] = \left[\frac{F'_q}{F'_v} \right] \left[\frac{F'_0}{F'} \right]$$

$$\text{NPQ can be estimated also as } \frac{F_M - F'_M}{F'_M} = \frac{k'_{NPQ} - k_{NPQ}}{k_{NPQ}}$$



Chlamydomonas reinhardtii

[klami] for intimists



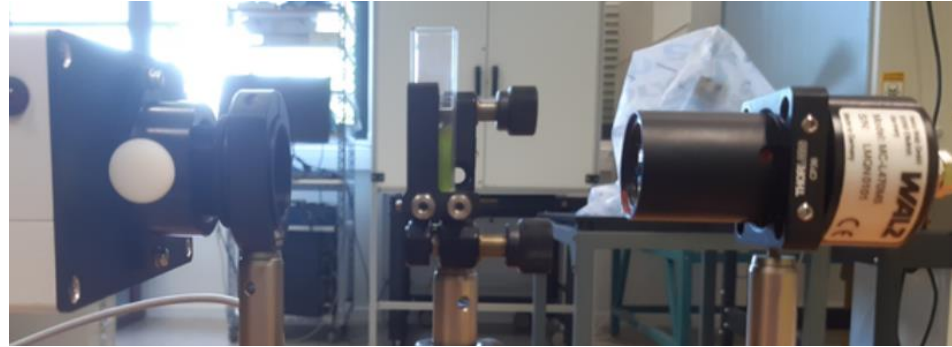
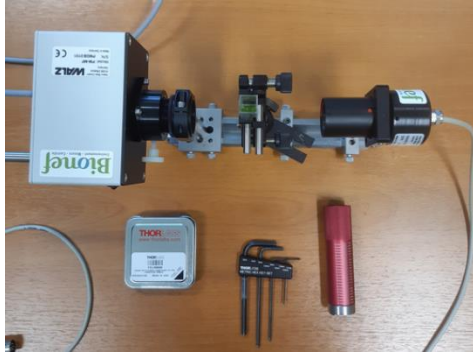
Swimming

Phototaxis

Photosynthesis

The Science Magazine of the Max Planck Society, 2014

N. Farès, Thèse, 2025



Microscopy PAM

Split in 2 groups
 ~1h30 with μ PAM
 ~1h30 with phytoPAM

PhytoPAM ED



Q&A